

$$F(k_x, k_y) = \int_{x_1}^{x_2} \int_{R \sin\left(\frac{2\pi x}{P}\right) - \frac{a}{2}}^{R \sin\left(\frac{2\pi x}{P}\right) + \frac{a}{2}} \sin(k_x x) \sin(k_y y) \, dx \, dy$$